

hw07

May 4, 2026

0.1 Problem 4 - The effect of Article VIII

```
[1]: import pandas as pd
import numpy as np
import statsmodels.api as sm

df = pd.read_csv("treaty.csv")
print(df.shape)
print(df.dtypes)
print(df.head())
print(df[["restrict", "art8", "imf_ccode"]].describe())
```

```
(4362, 48)
imf_ccode          int64
year              int64
art8              int64
restrict          int64
flexible          int64
gnpcap           float64
regnorm          float64
gdpgrow          float64
resgdp           float64
bopgdp           float64
useimfcr          int64
surveil          int64
univers          float64
resvol           float64
totvol           float64
tradedep         float64
lastrest          int64
lastrest2         int64
lastrest3         int64
military          float64
termlim           float64
parli            float64
shift_left        float64
leftshift_inc     float64
rightshift_inc    float64
leftshift_exc     float64
```

```

rightshift_exc      float64
leftshift_inc_leg   float64
rightshift_inc_leg   float64
leftshift_exc_leg    float64
rightshift_exc_leg   float64
shift_left_fixed     float64
leftrule            float64
l_leftshift_inc      float64
l_rightshift_inc      float64
l_leftshift_exc      float64
l_rightshift_exc      float64
l_leftshift_inc_leg   float64
l_rightshift_inc_leg  float64
l_leftshift_exc_leg   float64
l_rightshift_exc_leg  float64
year2                int64
year3                int64
interact1            float64
interact2            float64
interact3            float64
interact4            float64
interact5            float64

```

dtype: object

	imf_ccode	year	art8	restrict	flexible	gnpcap	regnorm	gdpgrow	\
0	1	1967	0	0	0	-4426.787840	0.0	21.839734	
1	1	1968	0	0	0	-1594.515902	0.0	31.462256	
2	1	1969	0	0	0	364.665761	0.0	-54.207744	
3	1	1970	0	0	0	195.271732	0.0	12.239170	
4	1	1971	0	0	0	-3361.533129	0.0	1.807442	

	resgdp	bopgdp	...	l_rightshift_inc_leg	l_leftshift_exc_leg	\
0	0.023025	-20.522696	...	NaN	NaN	
1	0.033197	11.044920	...	NaN	NaN	
2	0.029366	-7.629046	...	NaN	NaN	
3	0.028297	15.508446	...	NaN	NaN	
4	0.036925	-9.382553	...	NaN	NaN	

	l_rightshift_exc_leg	year2	year3	interact1	interact2	interact3	\
0	NaN	3869089	7610498048	NaN	NaN	NaN	
1	NaN	3873024	7622111232	NaN	NaN	NaN	
2	NaN	3876961	7633736192	NaN	NaN	NaN	
3	NaN	3880900	7645372928	NaN	NaN	NaN	
4	NaN	3884841	7657021440	NaN	NaN	NaN	

	interact4	interact5
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN

3	NaN	NaN
4	NaN	NaN

```
[5 rows x 48 columns]
```

	restrict	art8	imf_ccode
count	4362.000000	4362.000000	4362.000000
mean	0.543787	0.405089	76.808574
std	0.498136	0.490966	45.544833
min	0.000000	0.000000	1.000000
25%	0.000000	0.000000	39.000000
50%	1.000000	0.000000	75.000000
75%	1.000000	1.000000	114.000000
max	1.000000	1.000000	186.000000

The dataset contains 4,362 country-year observations across 48 variables. The outcome **restrict** and treatment **art8** are both binary, with 54% of observations recording a currency restriction and 41% having signed Article VIII. After dropping rows with missing covariate values the analysis sample is smaller. Which I will report after part a

Part (a)

```
[2]: # Cows given in question
covs = ["shift_left", "flexible", "regnorm", "gdpgrow",
        "resgdp", "bopgdp", "useimfcr", "surveil", "univers",
        "resvol", "totvol", "tradedep", "military", "termlim",
        "parli", "lastrest", "lastrest2", "lastrest3"]

cols = ["restrict", "art8", "imf_ccode"] + covs
df_model = df[cols].dropna()
print(f"Rows after dropping NAs: {len(df_model)} (dropped {len(df) -
    len(df_model)})")

X = sm.add_constant(df_model[["art8"] + covs])
y = df_model["restrict"]

logit_model = sm.Logit(y, X)
logit_result = logit_model.fit()
print(logit_result.summary2())

art8_coef = logit_result.params["art8"]
art8_se = logit_result.bse["art8"]
art8_pval = logit_result.pvalues["art8"]
print(f"\nart8 coefficient: {art8_coef:.4f}")
print(f"art8 SE: {art8_se:.4f}")
print(f"art8 p-value: {art8_pval:.4f}")
print(f"art8 odds ratio: {np.exp(art8_coef):.4f}")
```

Rows after dropping NAs: 4362 (dropped 0)
Optimization terminated successfully.

Current function value: 0.259201

Iterations 8

Results: Logit

```
=====
Model:                Logit                Method:                MLE
Dependent Variable: restrict                Pseudo R-squared: 0.624
Date:                2026-05-04 14:17 AIC:                2301.2731
No. Observations:    4362                BIC:                2428.8868
Df Model:            19                Log-Likelihood:    -1130.6
Df Residuals:        4342                LL-Null:            -3006.8
Converged:            1.0000                LLR p-value:        0.0000
No. Iterations:      8.0000                Scale:            1.0000
=====
```

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
const	0.2730	0.6013	0.4540	0.6498	-0.9055	1.4515
art8	-1.6956	0.1621	-10.4578	0.0000	-2.0133	-1.3778
shift_left	0.6228	0.2471	2.5207	0.0117	0.1385	1.1071
flexible	-0.5891	0.1465	-4.0209	0.0001	-0.8763	-0.3020
regnorm	0.0046	0.0024	1.9448	0.0518	-0.0000	0.0092
gdpgrow	-0.0101	0.0055	-1.8536	0.0638	-0.0208	0.0006
resgdp	-0.4410	0.3949	-1.1168	0.2641	-1.2150	0.3329
bopgdp	-0.0032	0.0053	-0.5911	0.5545	-0.0136	0.0073
useimfcr	0.9478	0.1282	7.3927	0.0000	0.6965	1.1991
surveil	0.4333	0.1584	2.7354	0.0062	0.1228	0.7438
univers	0.0362	0.0064	5.6909	0.0000	0.0238	0.0487
resvol	0.1399	0.1103	1.2676	0.2049	-0.0764	0.3562
totvol	0.3662	0.0967	3.7879	0.0002	0.1767	0.5556
tradedep	-0.0063	0.0013	-4.7190	0.0000	-0.0089	-0.0037
military	-0.3204	0.1400	-2.2882	0.0221	-0.5948	-0.0460
termlim	0.0484	0.1488	0.3251	0.7451	-0.2432	0.3399
parli	-0.0765	0.1528	-0.5005	0.6167	-0.3759	0.2230
lastrest	-1.6723	0.0746	-22.4061	0.0000	-1.8186	-1.5260
lastrest2	0.1345	0.0089	15.0656	0.0000	0.1170	0.1520
lastrest3	-0.0030	0.0003	-11.1741	0.0000	-0.0036	-0.0025

```
=====
```

art8 coefficient: -1.6956

art8 SE: 0.1621

art8 p-value: 0.0000

art8 odds ratio: 0.1835

```
[3]: print(np.exp(art8_coef))
```

0.18349751860998667

The logistic regression gets art8 coefficient of -1.696 (with SE = 0.162, $p < 0.001$), which corresponds to an odds ratio of 0.184. Intuitively, after adjusting for all covariates, countries that have

signed Article VIII have roughly 82% lower odds of imposing a currency restriction in a given year compared to non-signatories. It is important to note that this should not be interpreted as a causal effect. Countries self-select into signing Article VIII, so those that sign are likely already more committed to open capital markets, so the association may simply reflect this selection rather than any effect of the treaty itself. (Treatment is not random!!) Without a credible source of exogenous variation in treaty adoption, the strong negative association, no matter how statistically significant it is, cannot be given a causal interpretation.

Part (b)

```
[5]: logit_robust = logit_model.fit(
      cov_type="cluster",
      cov_kwds={"groups": df_model["imf_ccode"]}
    )
print(logit_robust.summary2())

art8_coef_r = logit_robust.params["art8"]
art8_se_r = logit_robust.bse["art8"]
art8_pval_r = logit_robust.pvalues["art8"]
print(f"\nart8 coefficient: {art8_coef_r:.4f}")
print(f"art8 cluster-robust SE: {art8_se_r:.4f}")
print(f"art8 p-value: {art8_pval_r:.4f}")
print(f"art8 odds ratio: {np.exp(art8_coef_r):.4f}")

print(f"\nSE comparison:")
print(f"Naive SE: {art8_se:.4f}")
print(f"Cluster-robust SE: {art8_se_r:.4f}")
```

Optimization terminated successfully.

Current function value: 0.259201

Iterations 8

Results: Logit

```
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Date:                2026-05-04 14:23 AIC:                2301.2731
No. Observations:    4362                BIC:                2428.8868
Df Model:            19                Log-Likelihood:    -1130.6
Df Residuals:        4342                LL-Null:            -3006.8
Converged:            1.0000                LLR p-value:        0.0000
No. Iterations:      8.0000                Scale:              1.0000
=====
```

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
const	0.2730	0.8336	0.3275	0.7433	-1.3608	1.9068
art8	-1.6956	0.2479	-6.8410	0.0000	-2.1813	-1.2098
shift_left	0.6228	0.2178	2.8601	0.0042	0.1960	1.0496
flexible	-0.5891	0.1847	-3.1890	0.0014	-0.9512	-0.2271
regnorm	0.0046	0.0037	1.2492	0.2116	-0.0026	0.0118

gdpgrow	-0.0101	0.0060	-1.6848	0.0920	-0.0219	0.0017
resgdp	-0.4410	0.7366	-0.5987	0.5494	-1.8848	1.0028
bopgdp	-0.0032	0.0051	-0.6206	0.5348	-0.0131	0.0068
useimfcr	0.9478	0.1684	5.6288	0.0000	0.6178	1.2778
surveil	0.4333	0.2137	2.0275	0.0426	0.0144	0.8522
univers	0.0362	0.0096	3.7926	0.0001	0.0175	0.0550
resvol	0.1399	0.1604	0.8719	0.3833	-0.1746	0.4543
totvol	0.3662	0.1192	3.0723	0.0021	0.1326	0.5997
tradedep	-0.0063	0.0018	-3.5895	0.0003	-0.0098	-0.0029
military	-0.3204	0.1515	-2.1151	0.0344	-0.6172	-0.0235
termlim	0.0484	0.1815	0.2665	0.7898	-0.3073	0.4040
parli	-0.0765	0.1655	-0.4619	0.6441	-0.4009	0.2480
lastrest	-1.6723	0.1041	-16.0574	0.0000	-1.8764	-1.4682
lastrest2	0.1345	0.0162	8.3160	0.0000	0.1028	0.1662
lastrest3	-0.0030	0.0006	-5.4125	0.0000	-0.0041	-0.0019

=====

art8 coefficient: -1.6956
 art8 cluster-robust SE: 0.2479
 art8 p-value: 0.0000
 art8 odds ratio: 0.1835

SE comparison:
 Naive SE: 0.1621
 Cluster-robust SE: 0.2479

Refitting the model with standard errors clustered by country (`imf_ccode`), the `art8` coefficient is unchanged, but the standard error increases from 0.162 to 0.248. This is expected as country-year observations within the same country are correlated over time, so the naive SE understates uncertainty by treating them as independent. But despite the larger SE, my conclusion does not change. The `art8` coefficient remains highly significant ($p < 0.001$), and the negative association between signing Article VIII and imposing currency restrictions is robust to clustering.

Part (c)

I disagree with this hypothetical suggestion. Ridge regression is most appropriate when the goal is prediction, where shrinkage reduces variance at the cost of introducing bias. Here I'd say the goal is more inference. Applying a ridge penalty would bias the `art8` coefficient toward zero, making it difficult to assess its true magnitude, and the resulting standard errors would no longer support valid hypothesis tests. Any penalized regression method, including Lasso as well (L1), shares this problem. Regularization is the wrong tool when the goal is unbiased inference on a specific coefficient rather than prediction.